

PS1 - IO

Tate Mason

```
# Packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(lubridate)
library(stargazer)
```

Please cite as:

Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
R package version 5.2.3. <https://CRAN.R-project.org/package=stargazer>

```
library(fixest)
library(here)
```

here() starts at /Users/tate/SchoolWork

```

library(readr)
library(tidyr)
library(stringr)

# Data
iri <- read_tsv("~/SchoolWork/Y2S1/I0/PS1/IRI.csv", show_col_types = FALSE)
names(iri) <- gsub("^t", "", names(iri)) # Clean names
iri <- iri %>%
  mutate(
    market_id = interaction(store_id, week_id, drop = TRUE),
    date_id = interaction(year, month, drop = TRUE),
    date = as.Date(paste(year, month, "01", sep = "-"))
  )

```

Question 1:

1.1

```

# Distinct counts per market
brands_per_market <- iri %>%
  distinct(market_id, brand) %>%
  count(market_id, name = "n_brands")

mfrs_per_market <- iri %>%
  distinct(market_id, parent) %>%
  count(market_id, name = "n_mfrs")

# Total market sales per market
total_sales_market <- iri %>%
  group_by(market_id) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Brand-level sales (market x brand)
brand_sales_market <- iri %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Per-market price & characteristics (means within market)

```

```

market_chars <- iri %>%
  group_by(market_id) %>%
  summarise(
    price_mean      = mean(price, na.rm = TRUE),
    fiber_mean      = mean(fiber, na.rm = TRUE),
    sugars_mean     = mean(sugars, na.rm = TRUE),
    flavored_mean   = mean(as.numeric(flavored), na.rm = TRUE),
    fortified_mean  = mean(as.numeric(fortified), na.rm = TRUE),
    .groups = "drop"
  )

# Helper to summarise numeric vectors
summarise_vec <- function(x) {
  tibble(
    mean  = mean(x, na.rm = TRUE),
    sd    = sd(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    max   = max(x, na.rm = TRUE)
  )
}

# Tables for 1.1
tbl_counts <- bind_cols(
  variable = c("n_brands", "n_mfrs"),
  bind_rows(
    summarise_vec(brands_per_market$n_brands),
    summarise_vec(mfrs_per_market$n_mfrs)
  )
)

tbl_total_sales <- bind_cols(
  variable = "total_sales (per market)",
  summarise_vec(total_sales_market$total_sales)
)

tbl_brand_sales <- bind_cols(
  variable = "brand_sales (per market-brand)",
  summarise_vec(brand_sales_market$brand_sales)
)

tbl_price_chars <- tribble(
  ~variable,                ~mean, ~sd, ~median, ~max,

```

```

"price (market mean)",
  summarise_vec(market_chars$price_mean)$mean,
  summarise_vec(market_chars$price_mean)$sd,
  summarise_vec(market_chars$price_mean)$median,
  summarise_vec(market_chars$price_mean)$max,
"fiber (market mean)",
  summarise_vec(market_chars$fiber_mean)$mean,
  summarise_vec(market_chars$fiber_mean)$sd,
  summarise_vec(market_chars$fiber_mean)$median,
  summarise_vec(market_chars$fiber_mean)$max,
"sugars (market mean)",
  summarise_vec(market_chars$sugars_mean)$mean,
  summarise_vec(market_chars$sugars_mean)$sd,
  summarise_vec(market_chars$sugars_mean)$median,
  summarise_vec(market_chars$sugars_mean)$max,
"flavored share (market mean)",
  summarise_vec(market_chars$flavored_mean)$mean,
  summarise_vec(market_chars$flavored_mean)$sd,
  summarise_vec(market_chars$flavored_mean)$median,
  summarise_vec(market_chars$flavored_mean)$max,
"fortified share (market mean)",
  summarise_vec(market_chars$fortified_mean)$mean,
  summarise_vec(market_chars$fortified_mean)$sd,
  summarise_vec(market_chars$fortified_mean)$median,
  summarise_vec(market_chars$fortified_mean)$max
)

summary_11 <- bind_rows(
  tbl_counts,
  tbl_total_sales,
  tbl_brand_sales,
  tbl_price_chars
)

summary_11

```

```

# A tibble: 9 x 5
  variable          mean      sd  median    max
  <chr>          <dbl>    <dbl>  <dbl>  <dbl>
1 n_brands      34.9      4.82    37     39
2 n_mfrs        3.95     0.259    4      4
3 total_sales (per market) 15555.  12358.  12557. 84567.

```

4 brand_sales (per market-brand)	446.	726.	238.	26470.
5 price (market mean)	0.235	0.0265	0.233	0.325
6 fiber (market mean)	1.30	0.0874	1.31	1.67
7 sugars (market mean)	8.31	0.205	8.33	10.2
8 flavored share (market mean)	0.640	0.0265	0.641	0.857
9 fortified share (market mean)	0.485	0.0341	0.487	0.667

```
# If you want a file:
# write_csv(summary_11, "summary_11_descriptives.csv")
```

1.2

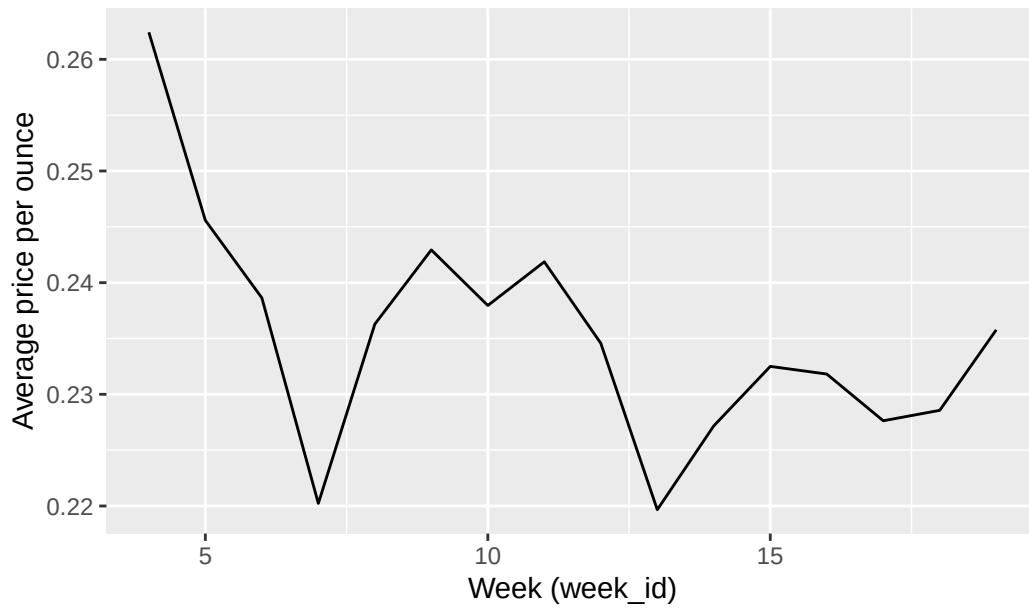
```
ts_simple <- iri %>%
  group_by(week_id) %>%
  summarise(
    avg_price = mean(price, na.rm = TRUE),
    avg_qty   = mean(quantity, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(week_id)

p_price <- ggplot(ts_simple, aes(x = week_id, y = avg_price)) +
  geom_line() +
  labs(title = "Average Cereal Price over Time",
       x = "Week (week_id)", y = "Average price per ounce")

p_qty <- ggplot(ts_simple, aes(x = week_id, y = avg_qty)) +
  geom_line() +
  labs(title = "Average Cereal Sales (Quantity) over Time",
       x = "Week (week_id)", y = "Average quantity (lbs)")

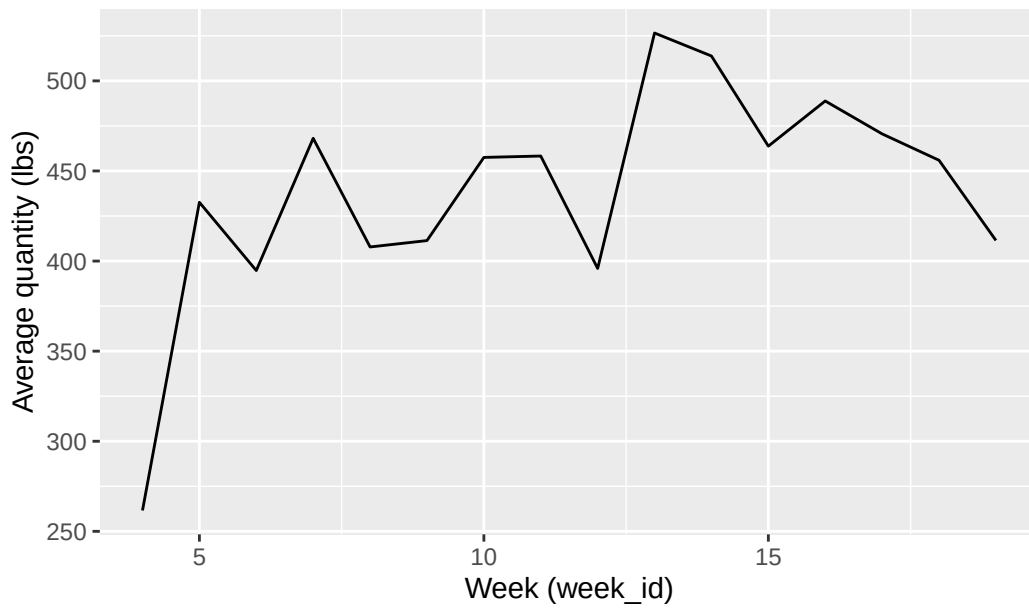
p_price
```

Average Cereal Price over Time



p_qty

Average Cereal Sales (Quantity) over Time



```
# Save figures
ggsave("avg_price.png", p_price, width = 10, height = 6, dpi = 300)
```

```

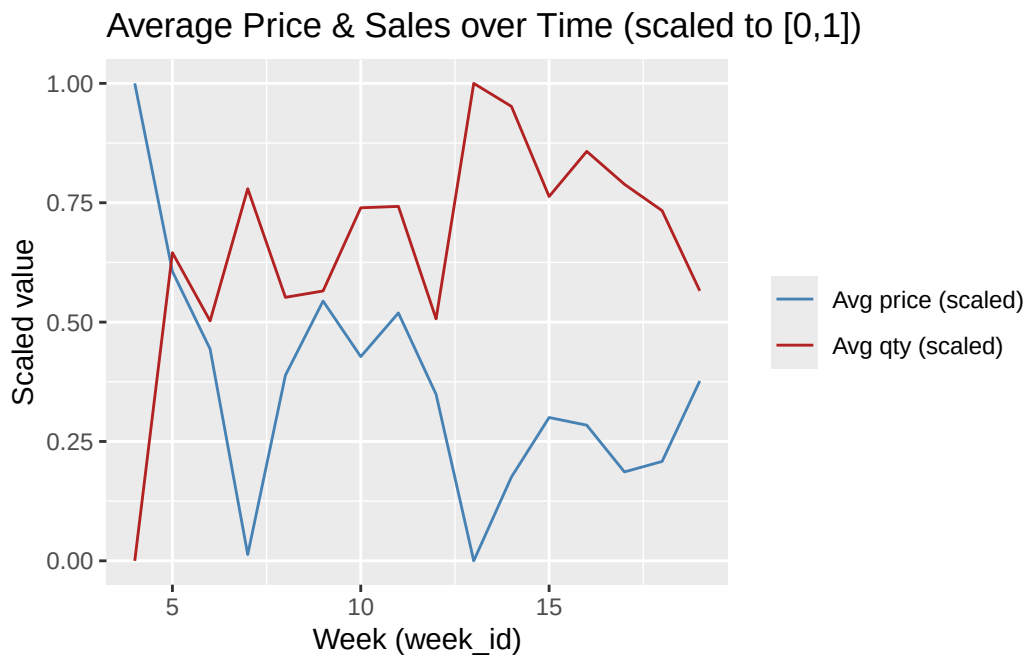
ggsave("avg_sales.png", p_qty, width = 10, height = 6, dpi = 300)

# Dual-axis style (scaled)
ts_long_scaled <- ts_simple %>%
  mutate(
    price_scaled = (avg_price - min(avg_price, na.rm = TRUE)) /
      (max(avg_price, na.rm = TRUE) - min(avg_price, na.rm = TRUE)),
    qty_scaled    = (avg_qty - min(avg_qty, na.rm = TRUE)) /
      (max(avg_qty, na.rm = TRUE) - min(avg_qty, na.rm = TRUE))
  ) %>%
  select(week_id, price_scaled, qty_scaled) %>%
  pivot_longer(-week_id, names_to = "series", values_to = "value")

p_dual <- ggplot(ts_long_scaled, aes(x = week_id, y = value, color = series)) +
  geom_line() +
  scale_color_manual(values = c("price_scaled" = "steelblue", "qty_scaled" = "firebrick"),
    labels = c("Avg price (scaled)", "Avg qty (scaled)")) +
  labs(title = "Average Price & Sales over Time (scaled to [0,1])",
    x = "Week (week_id)", y = "Scaled value", color = NULL)

p_dual

```



```
ggsave("avg_price_sales_dual.png", p_dual, width = 10, height = 6, dpi = 300)
```

1.3

```
# Brand HHI per market
brand_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
  mutate(share_brand = if_else(total_mkt_sales > 0, brand_sales / total_mkt_sales, 0))

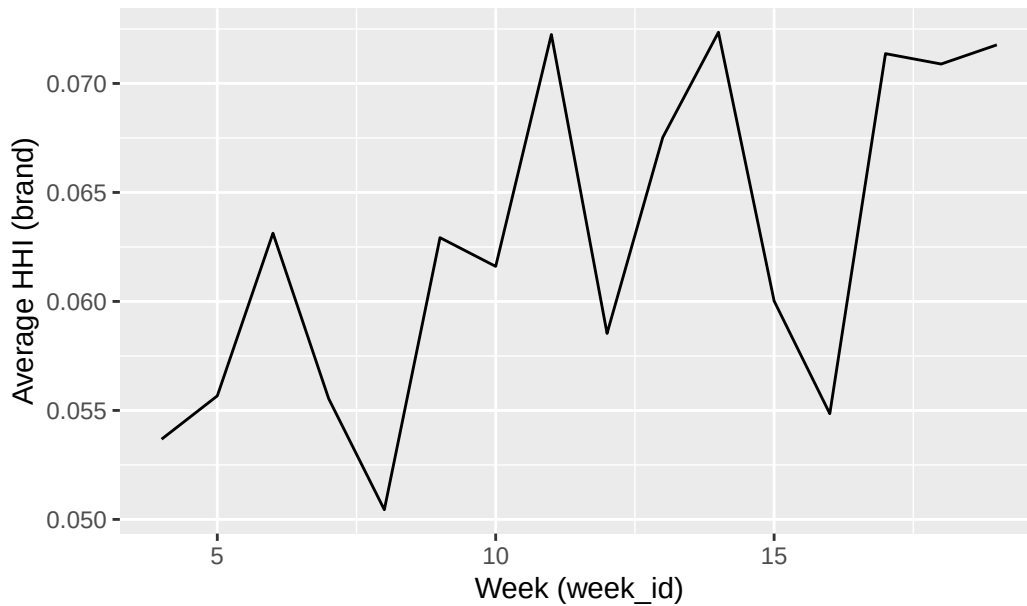
hhi_brand_market <- brand_shares %>%
  group_by(market_id) %>%
  summarise(HHI_brand = sum(share_brand^2, na.rm = TRUE), .groups = "drop")

hhi_brand_week <- iri %>%
  distinct(market_id, week_id) %>%
  inner_join(hhi_brand_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_brand = mean(HHI_brand, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)

p_hhi_brand <- ggplot(hhi_brand_week, aes(x = week_id, y = avg_HHI_brand)) +
  geom_line() +
  labs(title = "Average Brand-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (brand)")

p_hhi_brand
```


Average Brand-Level HHI Over Time



```
ggsave("HHI_brand.png", p_hhi_brand, width = 10, height = 6, dpi = 300)

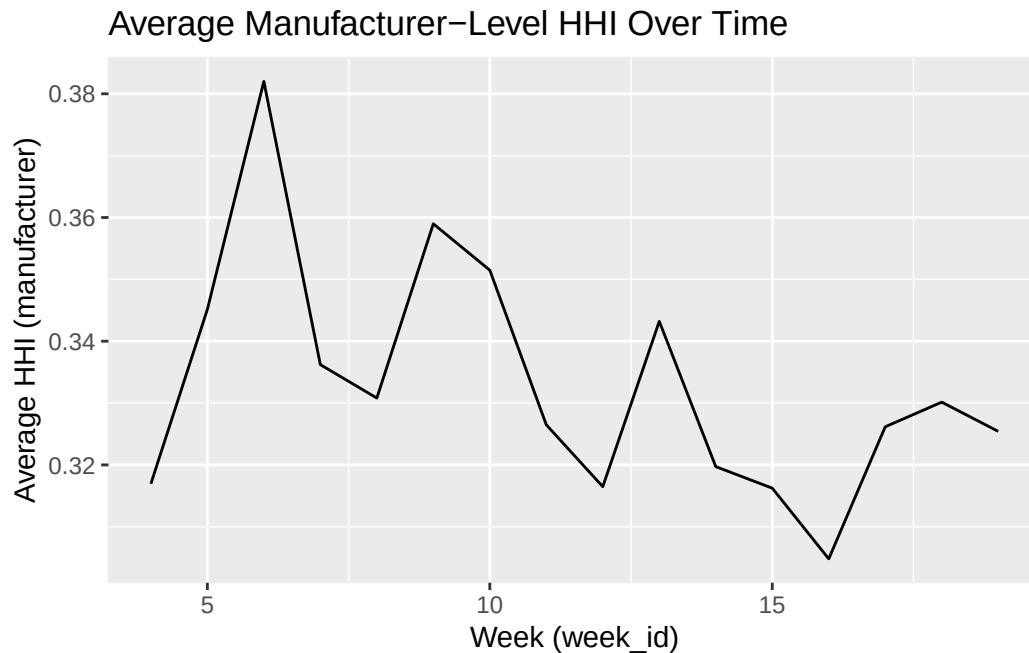
# Manufacturer (parent) HHI per market
parent_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, parent) %>%
  summarise(parent_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
  mutate(share_parent = if_else(total_mkt_sales > 0, parent_sales / total_mkt_sales, 0))

hhi_parent_market <- parent_shares %>%
  group_by(market_id) %>%
  summarise(HHI_parent = sum(share_parent^2, na.rm = TRUE), .groups = "drop")

hhi_parent_week <- iri %>%
  distinct(market_id, week_id) %>%
  inner_join(hhi_parent_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_parent = mean(HHI_parent, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)
```

```
p_hhi_parent <- ggplot(hhi_parent_week, aes(x = week_id, y = avg_HHI_parent)) +
  geom_line() +
  labs(title = "Average Manufacturer-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (manufacturer)")

p_hhi_parent
```



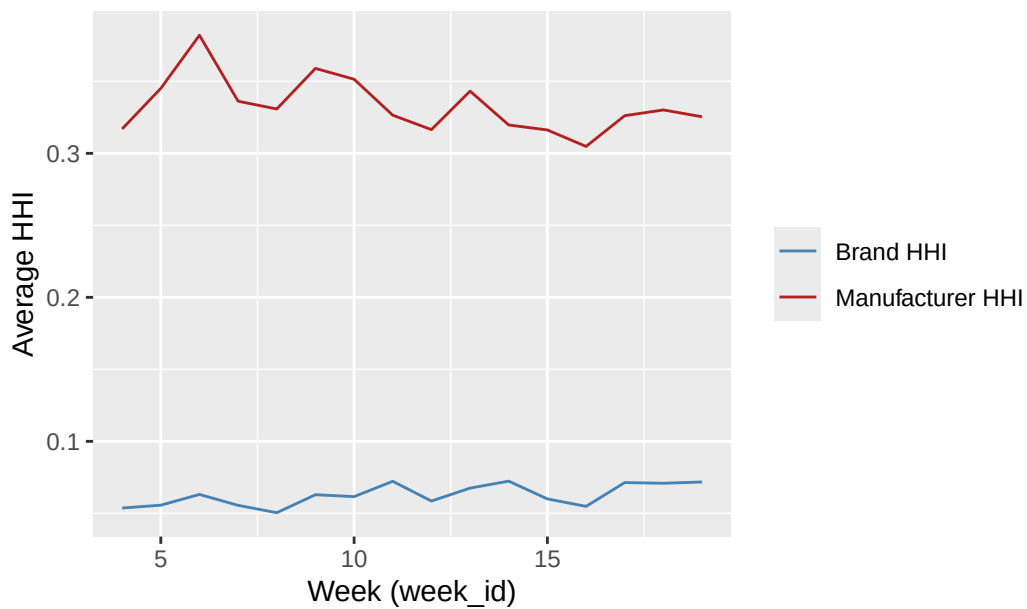
```
ggsave("HHI_parent.png", p_hhi_parent, width = 10, height = 6, dpi = 300)

# Combined comparison
hhi_both <- hhi_brand_week %>% inner_join(hhi_parent_week, by = "week_id")

p_hhi_both <- ggplot(hhi_both, aes(x = week_id)) +
  geom_line(aes(y = avg_HHI_brand, color = "Brand HHI")) +
  geom_line(aes(y = avg_HHI_parent, color = "Manufacturer HHI")) +
  scale_color_manual(values = c("Brand HHI" = "steelblue", "Manufacturer HHI" = "firebrick")) +
  labs(title = "Average HHI Over Time (Brand vs Manufacturer)",
       x = "Week (week_id)", y = "Average HHI", color = NULL)

p_hhi_both
```

Average HHI Over Time (Brand vs Manufacturer)



```
ggsave("HHI_brand_vs_parent.png", p_hhi_both, width = 10, height = 6, dpi = 300)
```

Question 2:

2.1

```
# Prepare data for regression
reg_data <- iri %>%
  group_by(market_id) %>%
  mutate(
    market_fe = as.factor(store_id),
    manufacturer_fe = as.factor(parent),
    time_fe = as.factor(week_id),
    brand_fe = as.factor(brand),
    p_jmt = price,
    total_sales = sum(quantity, na.rm = TRUE),
    sj = quantity / M,
    s0 = 1 - (total_sales / M),
    u = log(sj) - log(s0)
  ) %>%
  ungroup()
```

Our estimation model is thus:

$$u_{jmt} = \alpha p_{jmt} + \beta \mathbb{X}_{jmt} + \xi_{jmt}$$

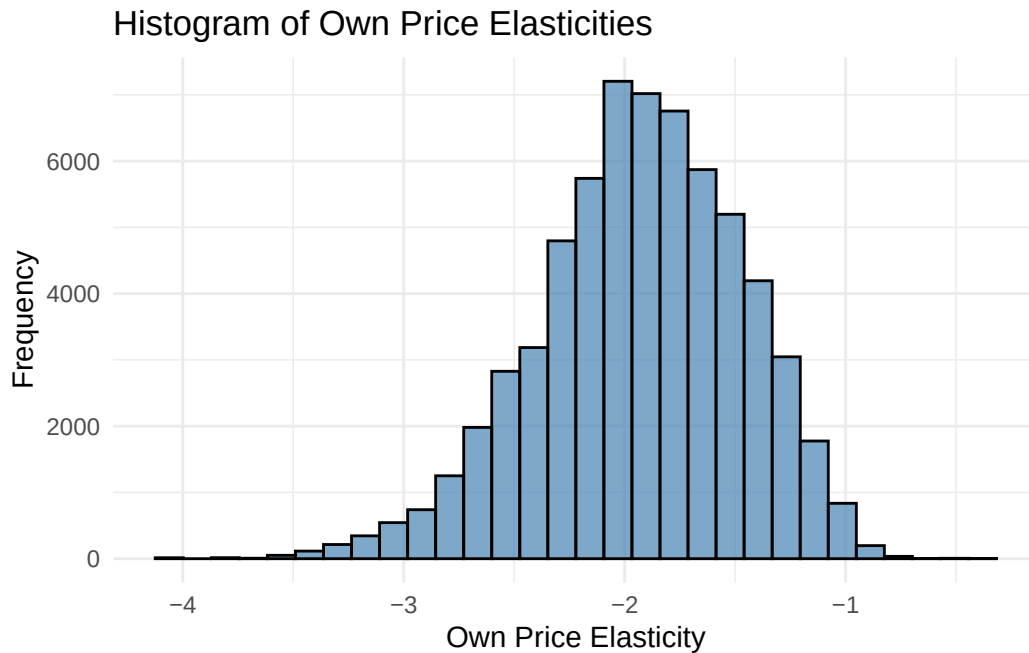
where $\mathbb{X}_{jmt}$ includes the product characteristics fiber, sugars, flavored, and fortified, as well as fixed effects for market, manufacturer, and time. and u_{jmt} is the log-odds of product j 's market share in market m at time t .

2.2

```
# Regression model
list <- c("fiber", "sugars", "flavored", "fortified",
         "market_fe", "manufacturer_fe", "time_fe")
reg_model <- feols(
  u ~ p_jmt + fiber + sugars + flavored + fortified |
  market_fe + manufacturer_fe + time_fe,
  data = reg_data
)

# Histogram of own price elasticities
reg_data <- reg_data %>%
  mutate(own_price_elasticity = coef(reg_model)["p_jmt"] * p_jmt * (1 - sj))

ggplot(reg_data, aes(x = own_price_elasticity)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities",
       x = "Own Price Elasticity", y = "Frequency") +
  theme_minimal()
```



2.3

```
# 2SLS with Price Instrument: Price*Quantity of Sugar

reg_data <- reg_data %>%
  mutate(
    z = sugars * sugar_price,
    z = log(z)
  )
stage2 <- feols(u ~ fiber + sugars + flavored + fortified |
  market_fe + manufacturer_fe + time_fe | p_jmt ~ z,
  data = reg_data)
summary(stage2)
```

```
TSLS estimation - Dep. Var.: u
                  Endo.    : p_jmt
                  Instr.    : z

Second stage: Dep. Var.: u
Observations: 63,966
Fixed-effects: market_fe: 642, manufacturer_fe: 4, time_fe: 16
```

Standard-errors: IID

	Estimate	Std. Error	t value	Pr(> t)
fit_p_jmt	-62.766445	7.898427	-7.94670	1.9468e-15 ***
fiber	-0.570755	0.066161	-8.62678	< 2.2e-16 ***
sugars	-0.120572	0.011575	-10.41695	< 2.2e-16 ***
flavored	0.179014	0.023118	7.74347	9.8166e-15 ***
fortified	0.377270	0.059710	6.31838	2.6607e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 2.49435 Adj. R2: 0.137296

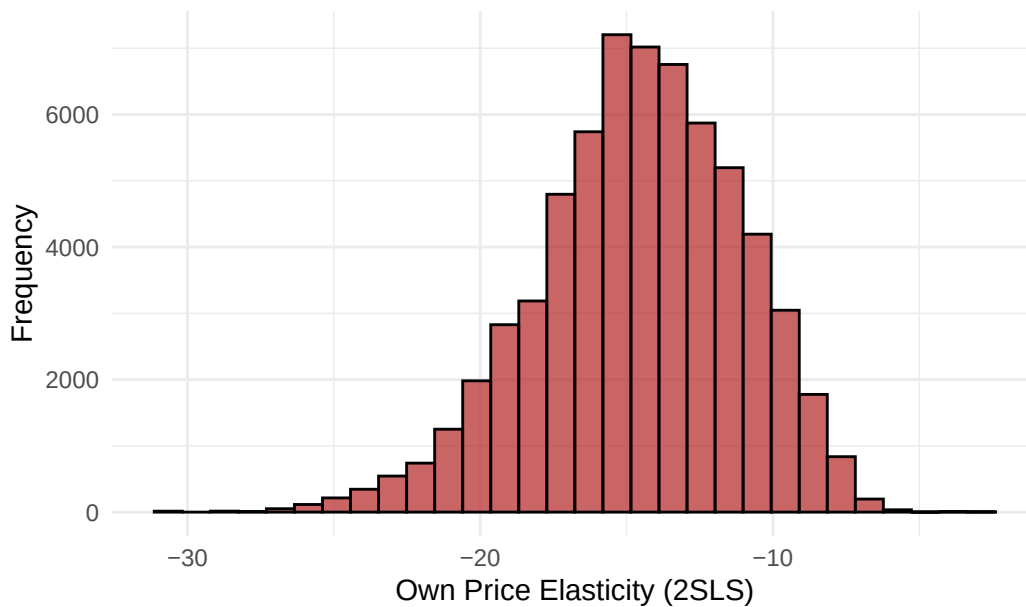
Within R2: 0.025342

F-test (1st stage), p_jmt: stat = 54.680, p = 1.436e-13, on 1 and 63,301 DoF.

Wu-Hausman: stat = 367.908, p < 2.2e-16 , on 1 and 63,300 DoF.

```
# Histogram of own price elasticities (2SLS)
reg_data <- reg_data %>%
  mutate(own_price_elasticity_2sls = coef(stage2)["fit_p_jmt"] * p_jmt * (1 - sj))
ggplot(reg_data, aes(x = own_price_elasticity_2sls)) +
  geom_histogram(bins = 30, fill = "firebrick", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities (2SLS)",
       x = "Own Price Elasticity (2SLS)", y = "Frequency") +
  theme_minimal()
```

Histogram of Own Price Elasticities (2SLS)



Using 2SLS results in estimates which are more negative than the OLS estimates, indicating that the OLS estimates were biased towards zero.

2.4

```
# Elasticity Matrix for Top 5 Brands by Sales
## Create average matrix by taking average of market level elasticity matrices

top_brands <- reg_data %>%
  group_by(brand) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(total_sales)) %>%
  slice_head(n = 5) %>%
  pull(brand)
elasticity_matrices <- list()
for (mkt in unique(reg_data$market_id)) {
  mkt_data <- reg_data %>% filter(market_id == mkt, brand %in% top_brands)
  n <- nrow(mkt_data)
  if (n < 5) next # Skip markets with fewer than 5 top brands
  elas_matrix <- matrix(0, nrow = n, ncol = n)
  rownames(elas_matrix) <- mkt_data$brand
  colnames(elas_matrix) <- mkt_data$brand
  for (i in 1:n) {
    for (j in 1:n) {
      if (i == j) {
        elas_matrix[i, j] <- coef(stage2)["fit_p_jmt"] * mkt_data$p_jmt[i] * (1 - mkt_data$sj[i])
      } else {
        elas_matrix[i, j] <- -coef(stage2)["fit_p_jmt"] * mkt_data$p_jmt[j] * mkt_data$sj[i]
      }
    }
  }
  elasticity_matrices[[as.character(mkt)]] <- elas_matrix
}
# Average elasticity matrix
avg_elasticity_matrix <- Reduce("+", elasticity_matrices) / length(elasticity_matrices)
avg_elasticity_matrix
```

	KELLOGGS FROSTED MINI WHEATS	
KELLOGGS FROSTED MINI WHEATS		-13.1423398
GENERAL MILLS CHEERIOS		0.2615167

GENERAL MILLS HONEY NUT CHEER	0.2654592
POST HONEY BUNCHES OF OATS	0.2657008
KELLOGGS FROSTED FLAKES	0.2576447
GENERAL MILLS CHEERIOS	
KELLOGGS FROSTED MINI WHEATS	0.2638177
GENERAL MILLS CHEERIOS	-13.2669948
GENERAL MILLS HONEY NUT CHEER	0.2671759
POST HONEY BUNCHES OF OATS	0.2681528
KELLOGGS FROSTED FLAKES	0.2604532
GENERAL MILLS HONEY NUT CHEER	
KELLOGGS FROSTED MINI WHEATS	0.2650948
GENERAL MILLS CHEERIOS	0.2639448
GENERAL MILLS HONEY NUT CHEER	-13.2314072
POST HONEY BUNCHES OF OATS	0.2691572
KELLOGGS FROSTED FLAKES	0.2576187
POST HONEY BUNCHES OF OATS	
KELLOGGS FROSTED MINI WHEATS	0.2623082
GENERAL MILLS CHEERIOS	0.2636535
GENERAL MILLS HONEY NUT CHEER	0.2664261
POST HONEY BUNCHES OF OATS	-13.2162577
KELLOGGS FROSTED FLAKES	0.2599669
KELLOGGS FROSTED FLAKES	
KELLOGGS FROSTED MINI WHEATS	0.2669329
GENERAL MILLS CHEERIOS	0.2632467
GENERAL MILLS HONEY NUT CHEER	0.2654629
POST HONEY BUNCHES OF OATS	0.2690599
KELLOGGS FROSTED FLAKES	-13.2852126

2.5

```
# Calculaete markups for each manufacturer
reg_data <- reg_data %>%
  mutate(
    own_price_elasticity_2sls = coef(stage2)["fit_p_jmt"] * p_jmt * (1 - sj),
    markup = -1 / own_price_elasticity_2sls
  )
markup_summary <- reg_data %>%
  group_by(parent) %>%
  summarise(
    avg_markup = mean(markup, na.rm = TRUE),
    sd_markup = sd(markup, na.rm = TRUE),
```



```

    median_markup = median(markup, na.rm = TRUE),
    max_markup = max(markup, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(avg_markup))
markup_summary

```

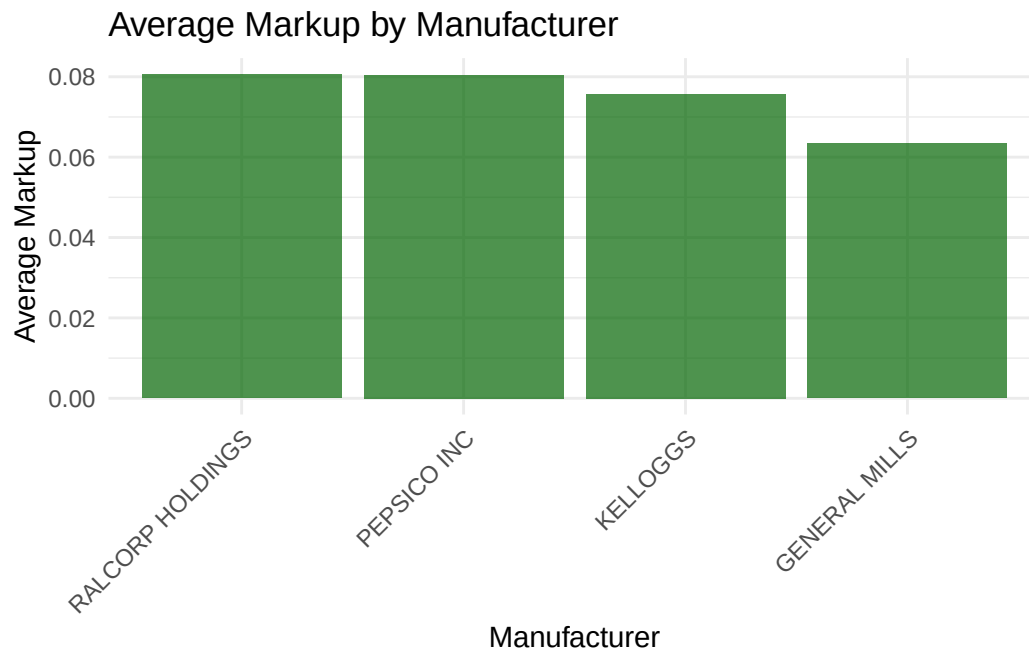
A tibble: 4 x 5

	parent <chr>	avg_markup <dbl>	sd_markup <dbl>	median_markup <dbl>	max_markup <dbl>
1	RALCORP HOLDINGS	0.0806	0.0225	0.0764	0.320
2	PEPSICO INC	0.0805	0.0179	0.0786	0.226
3	KELLOGGS	0.0757	0.0182	0.0727	0.182
4	GENERAL MILLS	0.0634	0.0127	0.0627	0.269

```

ggplot(markup_summary, aes(x = reorder(parent, -avg_markup), y = avg_markup)) +
  geom_bar(stat = "identity", fill = "darkgreen", alpha = 0.7) +
  labs(title = "Average Markup by Manufacturer",
       x = "Manufacturer", y = "Average Markup") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



2.6

```
# Nested by segment - construct data and calculate conditional shares
reg_data <- reg_data %>%
  mutate(
    nest = as.factor(segment)
  ) %>%
  group_by(market_id, nest) %>%
  mutate(
    total_nest_sales = sum(quantity, na.rm = TRUE),
    share_con = if_else(total_nest_sales > 0, quantity / total_nest_sales, 0)
  ) %>%
  ungroup()
```

The estimating equation is now:

$$u_{jmt} = \alpha p_{jmt} + \beta \mathbb{X}_{jmt} + \sigma_g s_{j|g} + \xi_{jmt}$$

where $s_{j|g}$ is the conditional share of product j within its nest g .

2.7

```
#2SLS with Price Instrument: Price*Quantity of Sugar and Conditional Share Instrument total p
reg_data <- reg_data %>%
  group_by(market_id) %>%
  mutate(
    z_cond = log(n_distinct(brand)),
    sjg = log(share_con),
    z = sugars * sugar_price,
    z = log(z)
  )
s2_nest <- feols(u ~ fiber + sugars + flavored + fortified |
  market_fe + manufacturer_fe + time_fe | p_jmt + sjg ~ z + z_cond,
  data = reg_data)
summary(s2_nest)
```

```
TSLS estimation - Dep. Var.: u
                  Endo.    : p_jmt, sjg
```

```

Instr.      : z, z_cond
Second stage: Dep. Var.: u
Observations: 63,966
Fixed-effects: market_fe: 642, manufacturer_fe: 4, time_fe: 16
Standard-errors: IID

```

	Estimate	Std. Error	t value	Pr(> t)
fit_p_jmt	-54.396707	6.652299	-8.17713	2.9602e-16 ***
fit_sjg	0.129922	0.078950	1.64563	9.9845e-02 .
fiber	-0.486850	0.059327	-8.20618	2.3258e-16 ***
sugars	-0.100596	0.012322	-8.16377	3.3063e-16 ***
flavored	0.174628	0.020285	8.60859	< 2.2e-16 ***
fortified	0.300395	0.055403	5.42196	5.9163e-08 ***

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.15952      Adj. R2: 0.137644
                Within R2: 0.025751
F-test (1st stage), p_jmt: stat = 32.803, p = 5.77e-15, on 2 and 63,300 DoF.
F-test (1st stage), sjg  : stat = 472.922, p < 2.2e-16 , on 2 and 63,300 DoF.
Wu-Hausman: stat = 942.470, p < 2.2e-16 , on 2 and 63,298 DoF.

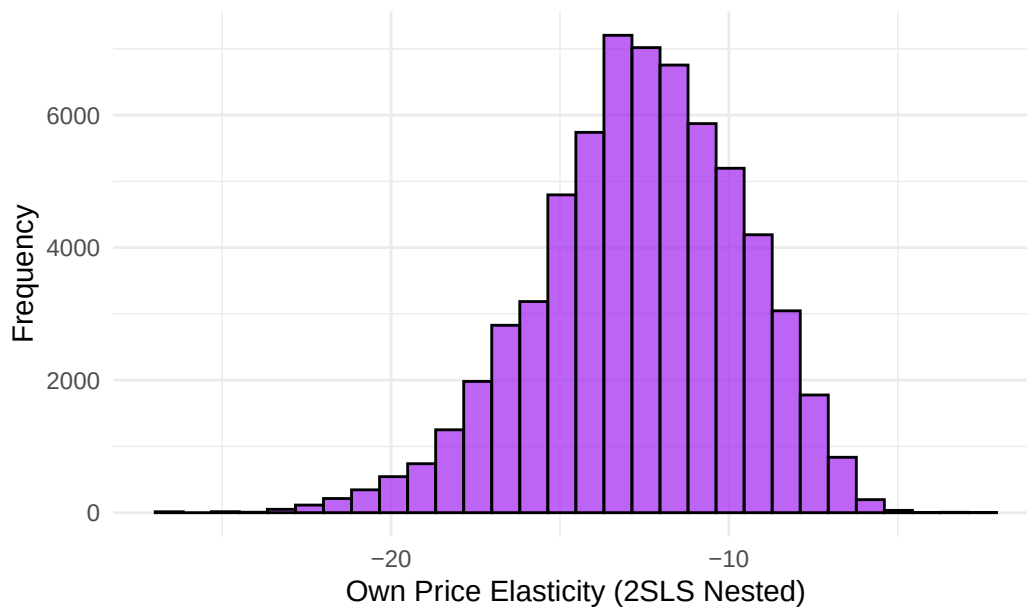
```

```

# Histogram of own price elasticities (2SLS Nested)
reg_data <- reg_data %>%
  mutate(own_price_elasticity_2sls_nest = coef(s2_nest)["fit_p_jmt"] * p_jmt * (1 - sj),
         sgj = log(coef(s2_nest)["fit_share_con"]))
ggplot(reg_data, aes(x = own_price_elasticity_2sls_nest)) +
  geom_histogram(bins = 30, fill = "purple", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities (2SLS Nested)",
       x = "Own Price Elasticity (2SLS Nested)", y = "Frequency") +
  theme_minimal()

```

Histogram of Own Price Elasticities (2SLS Nested)



2.8

```
# Elasticity Matrix for Top 5 Brands by Sales (Nested)
top_nest_brands <- reg_data %>%
  group_by(brand) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(total_sales)) %>%
  slice_head(n = 5) %>%
  pull(brand)
elasticity_nest_matrices <- list()
for (mkt in unique(reg_data$market_id)) {
  mkt_data <- reg_data %>% filter(market_id == mkt, brand %in% top_nest_brands)
  n <- nrow(mkt_data)
  if (n < 5) next # Skip markets with fewer than 5 top brands
  elas_matrix_nest <- matrix(0, nrow = n, ncol = n)
  rownames(elas_matrix_nest) <- mkt_data$brand
  colnames(elas_matrix_nest) <- mkt_data$brand
  for (i in 1:n) {
    for (j in 1:n) {
      if (i == j) {
        elas_matrix_nest[i, j] <- coef(s2_nest)["fit_p_jmt"] *
          mkt_data$p_jmt[i] * (1 / (1 - coef(s2_nest)["fit_sjg"])) *

```

```

        ((1 - coef(s2_nest)["fit_sjg"] * mkt_data$sjg[i]) - (1
        - coef(s2_nest)["fit_sjg"])) * mkt_data$sj[i])
    } else if (mkt_data$nest[i] == mkt_data$nest[j]) {
        elas_matrix_nest[i, j] <- -coef(s2_nest)["fit_p_jmt"] *
        mkt_data$p_jmt[i] * (1 / (1 - coef(s2_nest)["fit_sjg"])) *
        ((1 - coef(s2_nest)["fit_sjg"] * mkt_data$sjg[i]) - (1
        - coef(s2_nest)["fit_sjg"])) * mkt_data$sj[i])
    } else {
        elas_matrix_nest[i, j] <- coef(s2_nest)["fit_p_jmt"] *
        mkt_data$p_jmt[i] * (1 / (1 - coef(s2_nest)["fit_sjg"])) *
        ((1 - coef(s2_nest)["fit_sjg"] * mkt_data$sjg[i]) - (1 -
        coef(s2_nest)["fit_sjg"])) * mkt_data$sj[i])
    }
}
}
}
elasticity_nest_matrices[[as.character(mkt)]] <- elas_matrix_nest
}
# Average elasticity matrix (Nested)
avg_elasticity_nest_matrix <- Reduce("+", elasticity_nest_matrices) /
length(elasticity_nest_matrices)
avg_elasticity_nest_matrix

```

	KELLOGGS FROSTED MINI WHEATS	
KELLOGGS FROSTED MINI WHEATS		-16.726258
GENERAL MILLS CHEERIOS		3.934947
GENERAL MILLS HONEY NUT CHEER		3.884717
POST HONEY BUNCHES OF OATS		4.220221
KELLOGGS FROSTED FLAKES		3.412595
	GENERAL MILLS CHEERIOS	
KELLOGGS FROSTED MINI WHEATS	3.804315	
GENERAL MILLS CHEERIOS	-16.887474	
GENERAL MILLS HONEY NUT CHEER	4.515091	
POST HONEY BUNCHES OF OATS	4.865219	
KELLOGGS FROSTED FLAKES	4.027965	
	GENERAL MILLS HONEY NUT CHEER	
KELLOGGS FROSTED MINI WHEATS	3.769583	
GENERAL MILLS CHEERIOS	4.375811	
GENERAL MILLS HONEY NUT CHEER	-16.825855	
POST HONEY BUNCHES OF OATS	4.791045	
KELLOGGS FROSTED FLAKES	3.976238	
	POST HONEY BUNCHES OF OATS	

KELLOGGS FROSTED MINI WHEATS	4.271827
GENERAL MILLS CHEERIOS	5.006285
GENERAL MILLS HONEY NUT CHEER	4.941390
POST HONEY BUNCHES OF OATS	-16.782152
KELLOGGS FROSTED FLAKES	4.387227
	KELLOGGS FROSTED FLAKES
KELLOGGS FROSTED MINI WHEATS	3.286264
GENERAL MILLS CHEERIOS	3.794710
GENERAL MILLS HONEY NUT CHEER	3.807976
POST HONEY BUNCHES OF OATS	4.280453
KELLOGGS FROSTED FLAKES	-16.886185

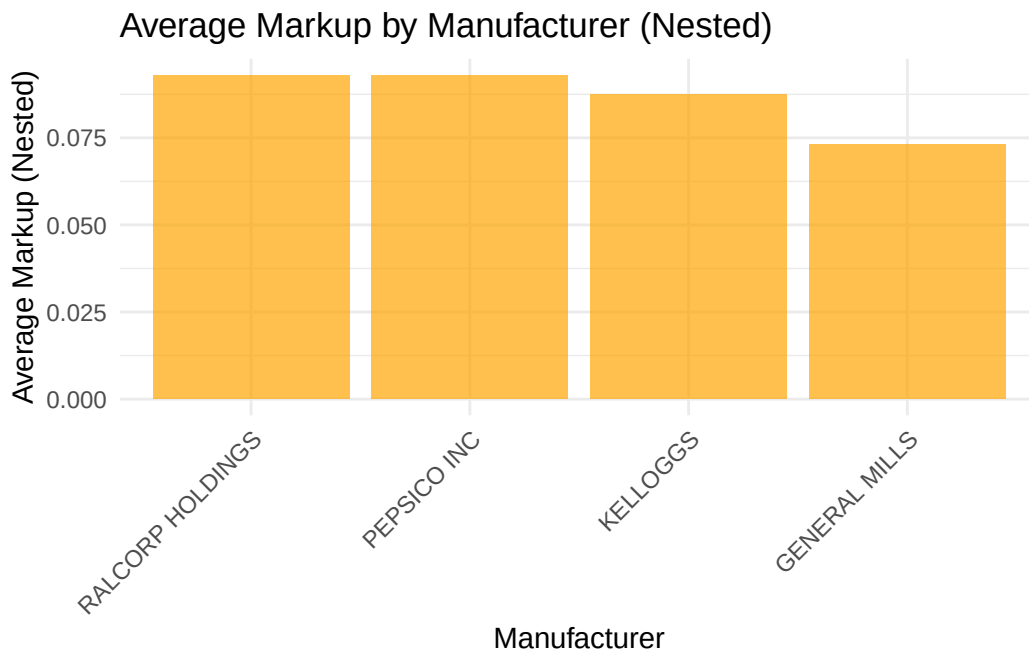
2.9

```
# Calculate average markups for each manufacturer (Nested)
reg_data <- reg_data %>%
  mutate(
    ope_nest = coef(s2_nest)["fit_p_jmt"] * p_jmt * (1 - sj),
    markup_nest = -1 / ope_nest
  )
markup_nest_summary <- reg_data %>%
  group_by(parent) %>%
  summarise(
    avg_markup_nest = mean(markup_nest, na.rm = TRUE),
    sd_markup_nest = sd(markup_nest, na.rm = TRUE),
    median_markup_nest = median(markup_nest, na.rm = TRUE),
    max_markup_nest = max(markup_nest, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(avg_markup_nest))
markup_nest_summary
```

A tibble: 4 x 5

	parent	avg_markup_nest	sd_markup_nest	median_markup_nest	max_markup_nest
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	RALCORP HOL~	0.0930	0.0259	0.0881	0.370
2	PEPSICO INC	0.0929	0.0206	0.0907	0.261
3	KELLOGGS	0.0873	0.0210	0.0839	0.210
4	GENERAL MIL~	0.0731	0.0146	0.0723	0.311

```
ggplot(markup_nest_summary, aes(x = reorder(parent, -avg_markup_nest),
                                y = avg_markup_nest)) +
  geom_bar(stat = "identity", fill = "orange", alpha = 0.7) +
  labs(title = "Average Markup by Manufacturer (Nested)",
       x = "Manufacturer", y = "Average Markup (Nested)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



2.10

```
# Cross - market diversion ratios between cheerios and corn flakes/froot loops
brands <- c("GENERAL MILLS CHEERIOS", "KELLOGGS CORN FLAKES", "KELLOGGS FROOT LOOPS")

row_stats <- reg_data %>%
  filter(brand %in% brands) %>%
  mutate(sj = pmin(sj, 0.999999)) %>%
  group_by(brand) %>%
  summarise(row_value = mean(-own_price_elasticity_2sls_nest * (sj / (1 - sj)), na.rm = TRUE),
            .groups = "drop")

diversion_matrix <- matrix(NA_real_, 3, 3, dimnames = list(brands, brands))
```

```

row_vals <- setNames(row_stats$row_value, row_stats$brand)

for (b in brands) {
  if (!is.na(row_vals[b])) diversion_matrix[b, setdiff(brands, b)] <- row_vals[b]
}

diversion_matrix

```

	GENERAL MILLS CHEERIOS	KELLOGGS CORN FLAKES	KELLOGGS FROOT LOOPS
GENERAL MILLS CHEERIOS	NA	0.23218833	
KELLOGGS CORN FLAKES	0.10098464	NA	
KELLOGGS FROOT LOOPS	0.08138226	0.08138226	

	KELLOGGS FROOT LOOPS
GENERAL MILLS CHEERIOS	0.2321883
KELLOGGS CORN FLAKES	0.1009846
KELLOGGS FROOT LOOPS	NA

2.11

```

# Find average markup for a hypothetical merger between General Mills and Kelloggs
merged_markup <- reg_data %>%
  mutate(
    parent_merged = if_else(parent %in% c("GENERAL MILLS", "KELLOGGS"), "GM_K", parent)
  ) %>%
  group_by(parent_merged) %>%
  summarise(
    avg_markup_merged = mean(markup_nest, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(avg_markup_merged))
merged_markup

```

```

# A tibble: 3 x 2
  parent_merged avg_markup_merged
  <chr>         <dbl>
1 RALCORP HOLDINGS 0.0930
2 PEPSICO INC      0.0929
3 GM_K            0.0799

```



```
# The average markup for the merged entity "GM_K" is:
merged_markup %>% filter(parent_merged == "GM_K") %>% pull(avg_markup_merged)
```

```
[1] 0.07994269
```

```
# This is higher than the individual average markups of General Mills and Kelloggs,
# indicating that the merger could lead to increased market power and higher markups
```

Question 3:

```
# Mixed Logit

library(tidyverse)
library(fixest)
library(BLPestimatorR)
library(readr)

# Load Data
iri <- read_tsv("~/SchoolWork/Y2S1/IO/PS1/IRI.csv", show_col_types = FALSE)

inc <- read_tsv("~/SchoolWork/Y2S1/IO/PS1/simulated_agents_income.csv", show_col_types = FALSE)

nkid <- read_tsv("~/SchoolWork/Y2S1/IO/PS1/simulated_agents_nchild.csv", show_col_types = FALSE)

# Mutate IRI data

iri <- iri %>%
  mutate(
    market_id = interaction(store_id, week_id, drop = TRUE),
    date_id = interaction(year, month, drop = TRUE),
    date = as.Date(paste(year, month, "01", sep = "-"))
  ) %>%
  filter(price > 0, quantity > 0) %>%
  group_by(market_id) %>%
  mutate(
    total_sales = sum(quantity, na.rm = TRUE),
    sj = quantity / M,
    s0 = 1 - (total_sales / M),
    u = log(sj) - log(s0)
```

```

) %>%
ungroup() %>%
mutate(
  pjmt = price
)

blp_instruments <- function(data) {
  data <- data %>%
    group_by(market_id) %>%
    mutate(
      BLP_sugar = (mean(sugars, na.rm = TRUE) - sugars)^2,
      BLP_fiber = (mean(fiber, na.rm = TRUE) - fiber)^2
    ) %>%
    ungroup()
}

iri <- blp_instruments(iri)

# Harmonize Income and nchild data
# Merge in income and number of kids data

inc_cols <- grep("^income\\.\\.\\.?\\d+$", names(inc), value = TRUE)
nkid_cols <- grep("^nchild\\.\\.\\.?\\d+$", names(nkid), value = TRUE)

inc_means <- inc %>%
  mutate(z_inc = rowMeans(across(all_of(inc_cols)), na.rm = TRUE)) %>%
  select(year, puma, z_inc)

nkid_means <- nkid %>%
  mutate(z_nchild = rowMeans(across(all_of(nkid_cols)), na.rm = TRUE)) %>%
  select(year, puma, z_nchild)
Rdraws <- length(inc_cols) # Number of draws

market_map <- iri %>%
  distinct(market_id, year, puma)

inc_mkt <- market_map %>%
  left_join(inc, by = c("year", "puma")) %>%
  select(market_id, all_of(inc_cols)) %>%
  rename(market_identifier = market_id)

nkid_mkt <- market_map %>%

```

```

left_join(nkid, by = c("year", "puma")) %>%
select(market_id, all_of(nkid_cols)) %>%
rename(market_identifier = market_id)

invinc_mkt <- inc_mkt
invinc_mkt[, inc_cols] <- lapply(inc_mkt[, inc_cols], function(x) 1 / x)

stdize_draw_cols <- function(df) {
  draw_names <- paste0("draw", seq_len(Rdraws))
  names(df)[-1] <- draw_names
  df
}
inc_mkt <- stdize_draw_cols(inc_mkt)
nkid_mkt <- stdize_draw_cols(nkid_mkt)
invinc_mkt <- stdize_draw_cols(invinc_mkt)

demo_draws <- list(
  inc = inc_mkt,
  nkid = nkid_mkt,
  invinc = invinc_mkt
)

iri <- iri %>%
  left_join(
    inc_means, by = c("year", "puma")) %>%
  left_join(
    nkid_means, by = c("year", "puma")
  ) %>%
  select(-puma, -year)
iri

```

A tibble: 63,966 x 28

	month	store_id	week_id	market_name	state	brand	parent	flavored	fortified
	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>
1	4	369	5	BOSTON	Massachus~	KELL~	KELLO~	1	0
2	4	108	5	BOSTON	Massachus~	GENE~	GENER~	1	1
3	4	374	5	BOSTON	Massachus~	KELL~	KELLO~	1	1
4	4	221	5	BOSTON	Massachus~	GENE~	GENER~	1	1
5	4	294	5	BOSTON	Massachus~	GENE~	GENER~	0	0
6	4	117	5	BOSTON	Massachus~	QUAK~	PEPSI~	1	0
7	4	378	5	BOSTON	Massachus~	GENE~	GENER~	0	0
8	4	310	5	BOSTON	Massachus~	GENE~	GENER~	1	0

```

9      4      237      5 BOSTON      Massachus~ GENE~ GENER~      1      1
10     4      321      5 BOSTON      Massachus~ KELL~ KELLO~      0      0
# i 63,956 more rows
# i 19 more variables: fiber <dbl>, sugars <dbl>, price <dbl>, quantity <dbl>,
#   sugar_price <dbl>, M <dbl>, segment <chr>, market_id <fct>, date_id <fct>,
#   date <date>, total_sales <dbl>, sj <dbl>, s0 <dbl>, u <dbl>, pjmt <dbl>,
#   BLP_sugar <dbl>, BLP_fiber <dbl>, z_inc <dbl>, z_nchild <dbl>

```

```

model <- as.formula( "sj ~ pjmt + fiber + sugars + flavored + fortified | fiber + sugars
+ flavored + fortified | pjmt + fiber + sugars | sugar_price + BLP_sugar + BLP_fiber +
fiber:z_inc + sugars:z_inc + fiber:z_nchild + sugars:z_nchild" )

```

```

prod_dat <- iri %>% select( market_id, sj, pjmt, fiber, brand, sugars, flavored, fortified,
sugar_price, BLP_sugar, BLP_fiber, z_inc, z_nchild )

```

```

mixed_dat <- BLP_data( model = model, market_identifier = "market_id", productData
= prod_dat, product_identifier = "brand", demographic_draws = demo_draws, integra-
tion_method = "MLHS", integration_accuracy = 1000, integration_seed = 123 ) theta2 <-
matrix(NA_real_, nrow = 3, ncol = 1 + length(demo_draws)) rownames(theta2) <- c("pjmt",
"fiber", "sugars") colnames(theta2) <- c("unobs_sd", names(demo_draws))

```

```

theta2["pjmt", "inv_income"] <- -1/mean(unlist(demographic_draws$inv_income[,1]),
na.rm = TRUE) # sensible start # fiber and sugars: estimate unobs sd and both demograph-
ics theta2["fiber", "unobs_sd"] <- 1 theta2["fiber", "income"] <- 0.1 theta2["fiber", "nchild"]
<- 0.1 theta2["sugars", "unobs_sd"] <- 1 theta2["sugars", "income"] <- 0.1 theta2["sugars",
"nchild"] <- 0.1

```

```

#----- # 8) Estimate #----- set.seed(123) blp_est <-
estimateBLP( blp_data = blp_data, par_theta2 = theta2, solver_method = "BFGS",
solver_maxit = 1000, solver_reltol = 1e-6, standardError = "heteroskedastic", extremum-
Check = FALSE, printLevel = 1 )

```

```

summary(blp_est)

```

```

#----- # 9) Own-price elasticities histogram #----- elas <-
get_elasticities(blp_est, "pjmt") # list of matrices, one per market own <- unlist(lapply(elas,
function(E) diag(E))) hist(own, breaks = 40, main = "Histogram of Own-Price Elasticities",
xlab = "Own-price elasticity")

```